

UNIVERSITY INSTITUTE OF ENGINEERING

COMPUTER SCIENCE & ENGINEERING

BACHELOR OF ENGINEERING

CSE/ 4th SEMESTER

OPERATING SYSTEM(24CST-205/24ITT-205)

Unit No.3

protection

Chapter No. 3

.Dr. Anu Kaushik

Lecture No.36

Topic :Goals, principles and domain of

Assistant Professor

Learning Objectives

- Understand the goals of protection, including ensuring system security, controlled access, and data integrity.
- Analyze the domain of protection, focusing on how access rights are defined and enforced for users, processes, and resources.

Goals of Protection

- In one protection model, computer consists of a collection of objects, hardware or software
- Each object has a unique name and can be accessed through a well-defined set of operations
- Protection problem - ensure that each object is accessed correctly and only by those processes that are allowed to do so

Principles of Protection

Guiding principle – principle of least privilege

- Programs, users and systems should be given just enough privileges to perform their tasks
- Limits damage if entity has a bug, gets abused
- Can be static (during life of system, during life of process)

Principles of Protection

- Or dynamic (changed by process as needed) – domain switching, privilege escalation
- “Need to know” a similar concept regarding access to data

Principles of Protection

Must consider “grain” aspect

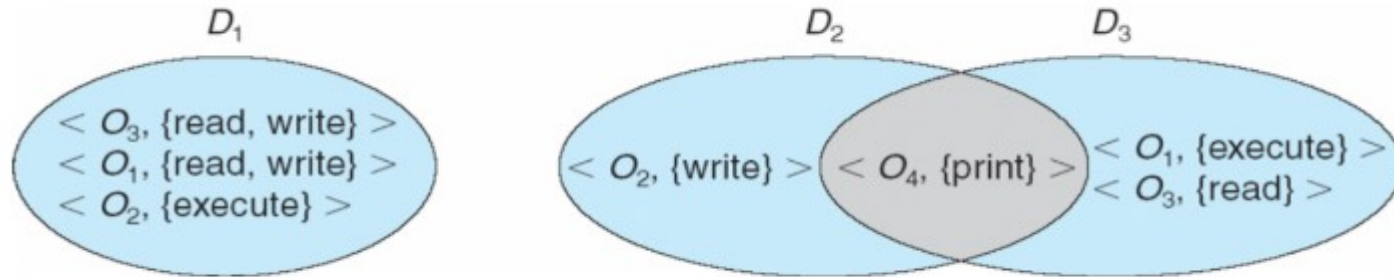
- Rough-grained privilege management easier, simpler, but least privilege now done in large chunks. For example, traditional Unix processes either have abilities of the associated user, or of root
- Fine-grained management more complex, more overhead, but more protective eg. File ACL lists, RBAC

Principles of Protection

- Domain can be user, process, procedure

Domain Structure

- Access-right = $\langle \text{object-name, rights-set} \rangle$ where rights-set is a subset of all valid operations that can be performed on the object
- Domain = set of access-rights





Summary of the Lecture

This lecture enables the students to understand the concept of protection in operating systems, including its goals, principles, and the domain of protection for secure resource access.



Next Lecture

- **Access Matrix**

Quiz/ FAQ's

Q1. Which principle ensures that a user/process gets only the minimum access required?

- a) Open access
- b) Least privilege
- c) Full control
- d) Sharing principle

Q2. In protection domain, a domain is defined as:

- a) A collection of users
- b) A set of access rights for objects
- c) A type of memory space
- d) A scheduling algorithm

References

<https://www.includehelp.com/c-programming-questions/>

<https://www.studytonight.com/operating-system/>

<https://computing.llnl.gov/tutorials/>

[https://www.tutorialspoint.com/operating_system/index.htm#:~:text=An%20operating%20system%20\(OS\)%20is,software%20in%20a%20computer%20system.](https://www.tutorialspoint.com/operating_system/index.htm#:~:text=An%20operating%20system%20(OS)%20is,software%20in%20a%20computer%20system.)

<https://www.javatpoint.com/os-tutorial>

<https://www.guru99.com/operating-system-tutorial.html>

<https://www.geeksforgeeks.org/operating-systems/>



Faculty-curated videos, NPTEL, Coursera, LinkedIn, or other relevant learning resources

- <https://nptel.ac.in/courses/106105214>
- <https://www.linkedin.com/learning/iot-foundations-operating-systems-fundamentals-2018>



Class-Wise Feedback

Faculty members are encouraged to generate a class specific QR code at the end of each session to collect brief student feedback. This can support self-reflection, improve teaching practices, and allow students to share insights, suggestions, or concerns anonymously.



Thank You

For queries

Email: anu.e12329@cumail.in